

# Generative AI and Foundation Models for Human Sensing (GenAI4HS)

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## Abstract

Until recently, progress in wearable and ubiquitous computing has been (broadly) driven by bespoke and specialized models, designed and developed for specific applications. This trend has now shifted, with rapidly increasing interest in developing generic, foundation models, or even incorporating existing Large Language Models (LLMs) into sensor pipelines. Although several early works have contributed to this shift, transformational changes have not yet been widely observed in the field. This workshop aims to provide a forum to discuss the underlying reasons for this, and for promoting and encouraging integration of foundation and Generative AI based models into wearables and ubiquitous applications. This workshop builds on the success of two previous iterations of the well-attended Ubicomp tutorials called “Solving the Activity Recognition Problem (SOAR)”. It brings together researchers from wide ranging fields into a discussion of the necessity and utility of these models, accompanied by a short tutorial exploring such integrations.

## CCS Concepts

• **Human-centered computing** → **Ubiquitous and mobile computing**; **Empirical studies in ubiquitous and mobile computing**; • **Computing methodologies** → *Machine learning*.

## Keywords

foundation models, generative ai, wearable and ubiquitous computing

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## 1 Introduction and Objective of the Workshop

Capturing and sensing human movements, behavior, and bio-signals via wearable and ubiquitous computing is critical for a number of applications, including fitness and health monitoring [1, 3], performing interactions [17], activity recognition [8, 16], and behavior modeling [4, 12]. As such, many of these sensors, e.g., wearables, have the capability to capture user behaviors in naturalistic, in-the-wild settings [2, 13], making them highly useful for real-world deployment scenarios.

Traditionally, owing to practical and resource constraints such as data scarcity and computational power, *single / small number of sensors* are utilized in such applications, to develop *bespoke and specialized models*. For example, Inertial Measurement Units (IMUs) are utilized for Human Activity Recognition (HAR), through the development of models such as DeepConvLSTM [14], which can *only perform HAR*. Critically, it is challenging to incorporate context information and world knowledge during training.

However, *times are changing* – datasets collected in recent years are not only *larger and more diverse*, but also contain *more sensing modalities*. For example, the large-scale Fenland study [11] collected both heart rate and accelerometry from more than 12000 participants, whereas the UK Biobank [2] has data from around 700k person days of data from ~95k participants. The Ego4d dataset [6] contains around 3700 hours of daily-life activity video, audio, detailed text descriptions, and a head-mounted IMU. Such datasets have enabled the development of *foundation models*, where the goal is to leverage large-scale data to develop *broadly usable, generic models*—often through self-supervised or cross-modal training—which can be *fine-tuned* to solve numerous downstream tasks and applications. A growing number of recent papers develop such foundation models, across diverse sensing modalities, e.g., IMU [18], PPG [15], video [19], and audio [7], involving both single and multiple modalities (i.e., uni- and multi-modal training). Especially, multi-modal foundation models enable the incorporation of complementary information from other modalities and generally result in performance improvements over unimodal training [].

More recently, there is increasing interest in *integrating Generative AI*, and more specifically Large-Language Models (LLMs), into human sensing applications which were previously modeled

solely using the sensing modalities. The promise of such an approach is the ability to *leverage world knowledge* present in these LLMs for the actual application. For instance, [10] demonstrates how sensor data can be fed to LLMs for predicting health outcomes. Similarly, LLMs have also been used as virtual annotators of activities of sensor streams [9]. Crucially, the use of LLMs for human sensing applications also offers *improved interpretability*, as the models verbalize and detail the chain of thoughts leading to specific outcomes [5]. As such, the utilization of LLMs is a growing trend, and represents another avenue for stepping away from specialized models to utilizing powerful, generic models on sensor data.

Building on the strong interest and success of two editions of tutorials in UbiComp<sup>1</sup>, we aim to *promote and encourage the development and integration of foundation models and Generative AI for human sensing applications*. It will contain a *mini-tutorial and mini-hackathon* discussing such approaches for sensor-based HAR in particular. We will also provide a forum for round-table discussions about these developments, and facilitate collaborations and networking opportunities for new and established researchers alike. Further, the keynote speech will provide insight into the area, and inspire novel research directions. Put together, the goal of the workshop is to bring together researchers tackling wide-ranging applications in wearable and ubiquitous computing, to discuss the still nascent (and very much underway) paradigm shift towards leveraging powerful models for human sensing applications.

## 2 Expected Attendance

Around 50 participants are expected to attend.

## 3 Topics

This workshop will explore novel techniques and challenges in human sensing and ubiquitous computing, with a focus on opportunities offered by Generative AI, foundation models, and multimodal data. The topics are organized into the following key areas:

### 3.1 Multi-Modal Learning for Human Sensing

**Multi-modal and Contrastive Representation Learning:** investigates learning shared, robust representations from multiple sensor modalities (e.g., inertial, visual, textual) using contrastive learning. Explores methods for learning representations from unlabeled (multi-modal) sensor data. Focuses on aligning representations to capture common information relevant to human state and activity while being robust to modality-specific noise.

**Mixture of Multi-Modal Experts:** explores using Mixture-of-Experts (MoE) models to combine specialized models, each trained on different human-centric modality combinations or specific tasks (e.g., activity recognition, emotion recognition, health monitoring).

**Foundation Models for Wearable and Ubiquitous Human Sensing:** investigates large, pre-trained models for sensor data specifically from wearable and ubiquitous devices monitoring human activity and physiology. Explores pre-training objectives for general-purpose representations, enabling efficient fine-tuning for diverse human sensing tasks.

**Augmenting Context Descriptions with Generative Models:** examines integrating knowledge from diverse sources (sensors, images, text) to create richer, holistic representations of people. Enables sophisticated context-aware applications via knowledge fusion, focusing on scenarios like personalized healthcare, smart homes, and assistive technologies.

**Large Language Model (LLM) Integration for Wearable and Ubiquitous Computing:** examines leveraging LLMs to enhance the holistic understanding of human behavior and interactions with users. Explores integrating knowledge from textual sources using LLMs, and enabling natural language interfaces, personalized feedback, and context-aware decision-making in applications like assistive technologies and smart environments. Also focuses on architectural approaches and technical challenges for incorporating LLMs directly into sensor data processing pipelines, enabling tasks like sensor data summarization, grounded reasoning of sensor data analysis, anomaly explanation, and natural language queries.

## 3.2 Adapting Multi-modal Models to Diverse Humans and Environments

**Transfer Learning and Domain Adaptation for Human Sensing:** addresses transferring knowledge from source domains/tasks with abundant data to target domains/tasks with limited data or different distributions, specifically in human-centric applications. Covers techniques for feature and model adaptation to new users, environments, and sensing modalities.

**Multi-Modal Domain Adaptation:** leverages information contained in different data domains to enable better knowledge transfer. Explores opportunities resulting from advancements in multimodal modeling techniques and relevant data collection efforts.

**Test-time Adaptation:** addresses model adaptation to distribution shifts encountered during deployment, without retraining.

**Simulation and Augmentation for Human-Centric Data:** Covers synthetic sensor data generation that mimics human behavior and physiology, including the use of foundation models for data augmentation and generation to create realistic and diverse datasets. This topic addresses the critical "sim2real" gap – the discrepancy between performance on simulated versus real-world data, and focuses on methods to improve the transferability from synthetic or augmented data to real-world, human-in-the-loop systems.

## 3.3 Deployment and Practical Considerations

**Real-World Use Cases in Foundational and Multi-Modal Human Sensing:** examines practical challenges in deploying human sensing systems for human-centric applications, including noisy data, sensor drift, missing data, individual variability, privacy concerns, and ethical considerations. Focuses on robust techniques and deployment strategies for areas like healthcare monitoring, smart homes, and personalized well-being.

## 4 Workshop Plan

We propose a one-day in-person workshop on **Sunday, October 12** with an invited talk, mini-tutorials, mini-hackathons and paper presentations separated into morning and afternoon sessions, round-table and panel discussions in the late afternoon. The workshop format is designed to emphasize collaboration between participants.

<sup>1</sup>Solving the Activity Recognition Problem (SOAR): <https://sites.google.com/view/soar-tutorial-ubicom2024/home>

## 4.1 Types of Submissions & Selection Criteria

Submissions will go through a single-phase review process with at least two reviewers. They will be reviewed and selected based on their originality, relevance, technical correctness, and potential for initiating a fruitful discussion at the workshop. Submissions with preliminary results or work-in-progress are particularly welcome.

## 4.2 Pre-workshop Activities

A Call-for-Participation will be advertised on a dedicated website for the workshop, and we invite submissions from all around the world. To maximize dissemination coverage, we will work with the UbiComp publicity and communication chairs. The papers will be presented in-person, and attendees will also be asked to fill in a survey regarding their accessibility requirements.

## 4.3 Workshop Activities

**4.3.1 Welcome (9:15-9:30).** The workshop begins with an introduction given by the organizers with a summary of the workshop's themes, goals, and outline.

**4.3.2 Keynote 1 (9:30-10:30).** Speaker to be confirmed.

**4.3.3 Paper Presentations 1 (10:30-11:30).** A set of 4-5 10-minute paper presentations.

**4.3.4 Mini-tutorial (11:30-12:30).** A session on multi-modal training and the utilization of foundational models for sensor-based HAR.

**4.3.5 Lunch and round table discussions (12:30-14:00).** Lunch break. Workshop organizers and invited mentors will spread around the workshop venue facilitating discussions on major and upcoming topics of the workshop with the attendees.

**4.3.6 Paper Presentations 2 (14:00-14:45).** A set of 3-4 10-minute paper presentations.

**4.3.7 Mini-hackathon (14:45-15:45).** A mini-hackathon on utilizing multi-modal large language models for sensor applications.

**4.3.8 Break (15:45-16:00).**

**4.3.9 Panel Discussion (16:00-16:45).** A panel discussion on the future of wearable and ubiquitous computing, where the organizers will actively engage with the audience to stimulate discussions.

**4.3.10 Closing Remarks (16:45-17:00).**

## 4.4 Post-Workshop Follow Up

The accepted papers will also appear in the UbiComp supplemental proceedings and the ACM Digital Library.

## 5 Organizers' Backgrounds

### 5.1 Harish Haresamudram

Harish Haresamudram is a final year PhD student in the School of Electrical and Computer Engineering at Georgia Institute of Technology. His research involves analyzing and modeling human movements and behaviors via the opportunistic use of wearable and ambient sensors, with a focus on improving health and well-being.

### 5.2 Chi Ian Tang

Chi Ian Tang is a Research Scientist at the Device Software Research group of Nokia Bell Labs, Cambridge. His research focuses on developing scalable and data-efficient solutions for mobile sensing. He is particularly interested in leveraging large quantities of unlabeled, real-world sensing data and multi-modal data to create robust and generalizable models.

### 5.3 Megha Thukral

Megha Thukral is a second-year Ph.D. student in the School of Interactive Computing at the Georgia Institute of Technology. Her research focuses on transfer learning and domain adaptation for human sensing, with the goal of building models that generalize across different users and environments.

### 5.4 Vitor Fortes Rey

Vitor Fortes Rey is a Senior Researcher at the German Research Center for Artificial Intelligence (DFKI). His research interests are representation learning for Human Activity Recognition, especially using wearable sensors. He is also interested in reducing labeled data through simulation and multi-modal representation learning.

### 5.5 Sungho Suh

Sungho Suh is a Tenure-Track Assistant Professor at Korea University, Republic of Korea. His research interests are multi-modal learning and generative models, focusing on industrial applications.

### 5.6 Bo Zhou

Bo Zhou is Deputy Head of the Embedded Intelligence Department at DFKI. His research focuses on advancing machine learning – particularly foundation models, transfer learning with 3D simulation, and edge AI – by integrating multi-modal sensor data, optimizing hardware-AI codesign for efficiency.

### 5.7 Paul Lukowicz

Paul Lukowicz is both Scientific Director of the Embedded Intelligence Department at DFKI and Professor of Computer Science at RPTU in Germany since 2012, where he heads the Embedded Intelligence group. His research focuses on context-aware ubiquitous and wearable systems with applications in areas ranging from healthcare through industry 4.0 to smart cities.

### 5.8 Thomas Plötz

Thomas Ploetz is currently a Professor at the School of Interactive Computing at Georgia Institute of Technology. His research agenda focuses on applied machine learning, that is developing systems and innovative sensor data analysis methods for real world applications. Primary application domain for his work is computational behavior analysis where he develops methods for automated and objective behavior assessments in naturalistic environments.

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## A Call for Participation

Human sensing plays a critical role in wide ranging applications such as human activity recognition, health monitoring and behavior analysis, often driven by the development of *specialized models*. Meanwhile, broadly usable, generic models, particularly involving Generative AI and foundational models, have brought transformative advancements in other domains. Yet, similar breakthroughs have not yet been widely observed in wearable and ubiquitous computing.

This workshop aims to bring researchers and practitioners to discuss the recent trends and challenges in building foundational models for human sensing, and integrating advances from Generative AI into sensor pipelines. The workshop will present an opportunity for researchers to present their latest findings, acquire hands-on experience, and discuss the future of the field. It will also contain a mini-tutorial and mini-hackathon on using Generative AI and foundational models for human sensing applications and host presentations for the accepted submissions, which will take the form of full and position papers. We particularly welcome preliminary or work-in-progress results, in cover topics such as, but not limited to:

- Foundational models for wearable and ubiquitous computing.
- Augmenting context descriptions with generative models.
- Multi-modal domain adaptation or representation Learning for wearable and ubiquitous computing.
- Using generative models for data simulation, e.g., text-to-motion, diffusion models, and others. This also includes physical simulation and addressing the critical "sim2real" gap.
- Large Language Model (LLM) integration for ubiquitous computing.
- Challenges in practical deployment of Generative AI and foundational models in-the-wild.

Submissions will go through a single-phase review process with at least 2 reviewers. They will be reviewed and selected based on their originality, relevance, technical correctness, and their potential for initiating fruitful discussions at the workshop. Note, that position papers are not expected to present finished research projects. We rather ask for thought-provoking ideas or initial explorations of a topic. At the workshop, accepted submissions will be presented in a 10-minute presentation. At least one author of an accepted submission must attend the workshop. All accepted workshop papers can be included in the ACM Digital Library (DL) and the supplemental proceedings.